



Mathematics Extension 1

YEAR 12 TRIAL EXAMINATION 2008

*Time allowed : Reading time – 5 minutes
Working time - 2 hours*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- Write using blue or black pen.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- **Each question is to be started on a new page** and you are to write your name and teacher's name on each page.
- The marks allocated for each question are indicated

Name : _____ Class Teacher : _____

Q	1	2	3	4	5	6	7	Total
Mark	/12	/12	/12	/12	/12	/12	/12	/84

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Total marks – 84

Attempt Questions 1 – 7.

All questions are of equal value.

Start each question on a new page. Extra writing paper is available.

Question 1 (12 marks) Start a new page

Marks

(a) Find $\int \frac{1}{x^2+3} dx$ 2

(b) Ten people sit around a round table. How many arrangements are possible if three particular people want to sit together? 2

(c) Find to the nearest degree the size of the acute angle between the lines 2

$$\begin{aligned}x + 2y - 1 &= 0 \\ 3x - 2y + 4 &= 0\end{aligned}$$

(d) If $y = \frac{3x+5}{x-4}$ 2

Express the inverse function as a function of x

(e) For the points A (-5, 2) and B(2, 0)

(i) Write down the coordinates of P, the point that divides AB internally in the ratio $k:1$ 2

(ii) If P lies on $xy=1$, show that $k^2 - 2k + 11 = 0$ 2

Question 2 (12 marks) Start a new page

Marks

- (a) Given that a root of the equation $e^x + e^{-x} - 3 = 0$ is close to 1, use one application of Newton's Method of Approximation to find a second approximation to this root (correct to 2 decimal places)

2

- (b) Prove by Mathematical induction that $2^{2n+1} + 1$ is divisible by 3 for all non-negative integral values of n .

3

- (c) (i) Express $\sin x - \cos x$ in the form $A \sin(x - \alpha)$, with $A > 0$ and $0 < \alpha < \frac{\pi}{2}$

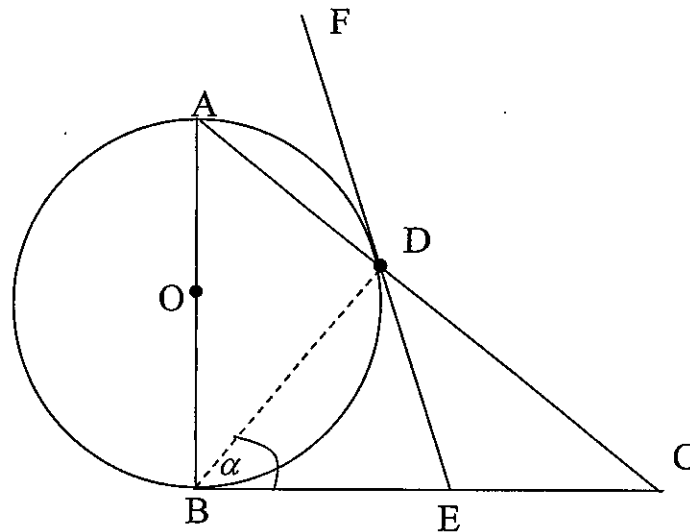
2

- (ii) Hence or otherwise determine $\lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{\sin x - \cos x}{x - \frac{\pi}{4}} \right)$

1

(d)

Not to scale



In the diagram, AB is a diameter of the circle, centre O, and BC is tangential to the circle at B. The line ADC intersects the circle at D. The tangent to the circle at D intersects BC at E. Let $\angle EBD = \alpha$

- (i) Copy the diagram onto your page

- (ii) Prove that $\angle EDC = \frac{\pi}{2} - \alpha$

4

Question 3 (12 marks) Start a new page**Marks**(a) Consider the function $f(x) = \cos^{-1} x$ (i) Sketch the graph of $y = f(x)$, stating clearly its range and domain. 3(ii) Find the volume of the solid formed by rotating the arc of the curve $y = \cos^{-1} x$ that is in the positive quadrant about the y axis. 2(b) Evaluate $\int_0^{\frac{\pi}{2}} [(3 \sin x) - 1]^2 \cos x \, dx$ using the substitution $u = (3 \sin x) - 1$ 3(c) (i) If $t = \tan \frac{\theta}{2}$, write down expressions for $\sin \theta$ and $\cos \theta$ in terms of t 1(ii) Hence or otherwise solve the equation 3

$$\sqrt{3} \sin \theta = 1 + \cos \theta \quad \text{in the domain } 0 \leq \theta \leq 2\pi$$

Question 4 (12 marks) Start a new page

Marks

(a)

(i) Show that the derivative of $x \tan x - \ln(\cos x)^{-1}$ is $x \sec^2 x$ **2**

(ii) Hence or otherwise evaluate $\int_0^{\frac{\pi}{4}} x \sec^2 x \, dx$ **2**

(b) A team of 6 girls is to be chosen from 10 girls.

(i) Find the number of ways that two particular girls, A and B, are **both included**. **1**

(ii) Find the number of ways that A and B are **both excluded**. **1**

(iii) Find the probability that either A is included or B is included, but A and B are not included together. **2**

(c) A particle is released from rest at the origin on a straight line, when x metres from the origin, its acceleration is given by

$$\frac{18}{(x-4)^2} \text{ m/s}^2, \text{ for } x < 4.$$

(i) In which direction will the particle first move? **1**

(ii) Find the particle's velocity when it reaches $x = 2$ **3**

Question 5 (12 marks) Start a new page

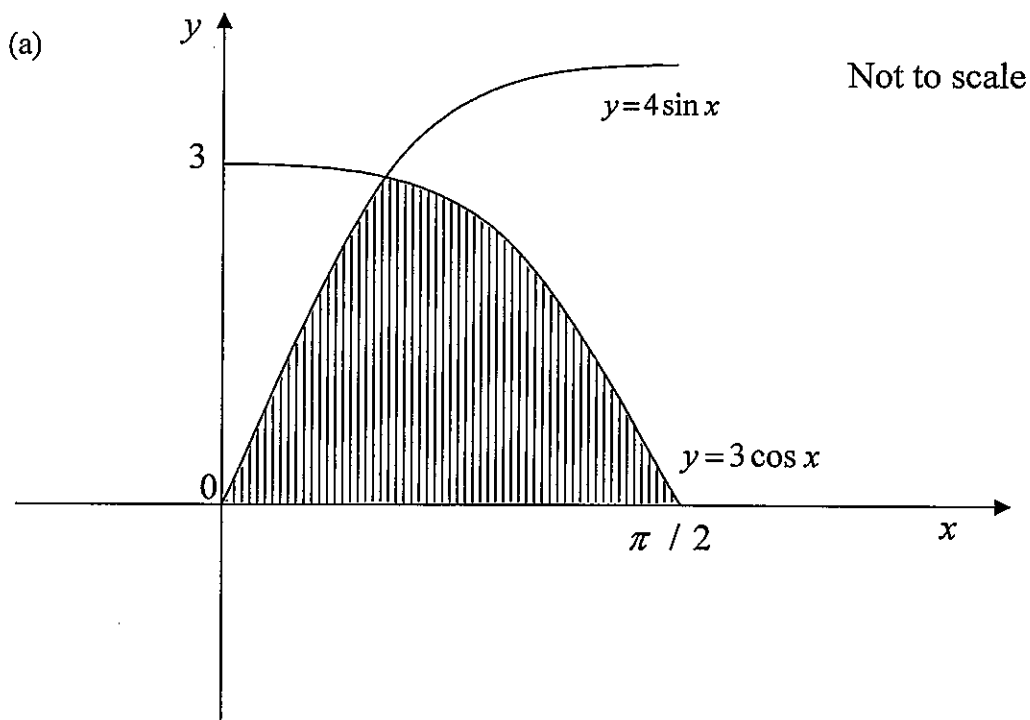
- (a) A particle moves along in a straight line such that its displacement x metres from an origin O at time t seconds is given by:

$$x = 4 \sin \frac{\pi}{2} t$$

- (i) Show that this motion is simple harmonic motion. 3
- (ii) State the amplitude and the period of this motion. 2
- (iii) Calculate the maximum speed attained by the particle. 3
- (b)
- Use long division to divide the polynomial $f(x) = x^4 - x^3 + x^2 - x + 1$ by the polynomial $g(x) = x^2 - 3$. 2
- Express your answer in the form $f(x) = g(x).q(x) + r(x)$
- (c) Differentiate $\cos^{-1}(e^{-x})$ with respect to x , putting your answer in simplest form. 2

Question 6 (12 marks) Start a new page

Marks



The diagram shows the graphs of $y = 4 \sin x$ and $y = 3 \cos x$. Show that the area of the shaded region in the diagram is 2 units^2 .

3

- (b) A particle projected from ground level at an angle θ to the horizontal has its position at time t given by the coordinates

$$x = Vt \cos \theta, \quad y = Vt \sin \theta - \frac{1}{2} gt^2 \quad (\text{DO NOT PROVE THESE})$$

- (i) Using these equations, find the maximum height reached in terms of V , g and θ .
- (ii) What is the speed of the object at its maximum height?

2

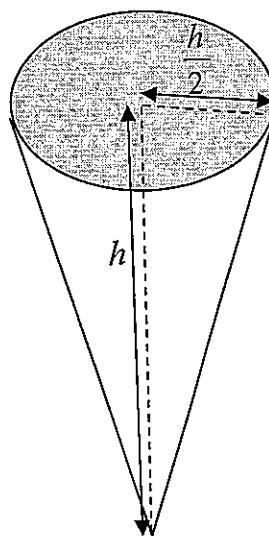
1

Question 6 continues on page 9

(c)

Not to scale

A cone has a depth of h and radius of $\frac{h}{2}$, as shown.



(i) Show that the volume of the cone is given by $V = \frac{1}{12} \pi h^3$ 1

(ii) Water is poured in at a rate of $10 \text{ mm}^3 / \text{s}$. Find the rate at which the depth, $h \text{ mm}$, is increasing when the depth of water in the cone is 50 mm . 2

The cone is filled to a depth of 100 mm and pouring then stops. A hole is then opened at the vertex of the cone and water flows out at a rate of $\pi h^2 \text{ mm}^3 / \text{s}$

(iii) Find $\frac{dh}{dt}$ in mm/s 2

(iv) Hence find how long it takes to empty the cone. 1

End of Question 6

Question 7 (12 marks) Start a new page

- (a) The mass, M , of a radioactive element decreases at a rate proportional to the mass,

ie. $\frac{dM}{dt} = -kM$ where k is a constant.

- (i) Show that the function $M = M_0 e^{-kt}$, where M_0 is the initial mass, provides such a rate. 1

- (ii) The “half-life” period of an element is the time taken for any given mass to be reduced by half. If the half-life period is T , 3

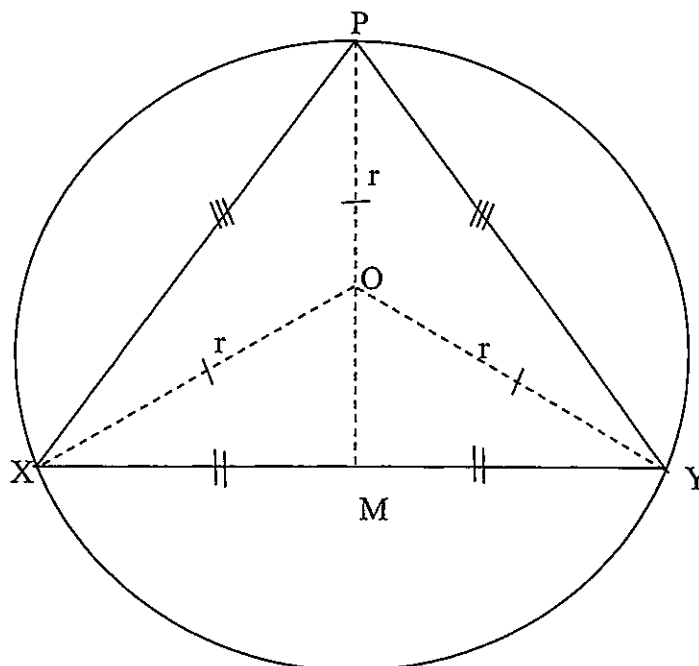
prove that $k = \frac{\ln 2}{T}$

Question 7 continues on page 11

(b)

Marks

Not to scale



In the diagram above, O is the centre of a circle of constant radius r . A variable chord XY subtends an angle 2θ at the centre O . Let P be the point on the major arc XY so that $\triangle XPY$ is isosceles with $XP = YP$. Let PO produced meet XY at M so that PM is the perpendicular bisector of the chord XY .

- (i) Prove that the area, A , of $\triangle XPY$ is given by:

3

$$A = r^2 \sin \theta (1 + \cos \theta)$$

- (ii) Show that $\frac{dA}{d\theta} = r^2 (2\cos^2 \theta + \cos \theta - 1)$

2

- (iii) Show that $\triangle XPY$ has maximum area when it is an equilateral triangle. You may assume it is a maximum – **you are NOT**

3

required to test $\frac{d^2 A}{d\theta^2}$

End of paper

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

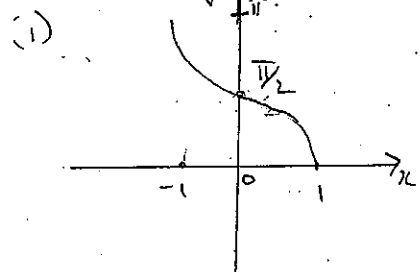
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

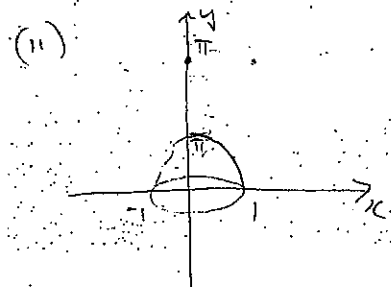
Question 3.

a) $f(x) = \cos^{-1} x$



D: $-1 \leq x \leq 1$

R: $0 \leq y \leq \pi$



$$V = \pi \int_0^{\pi/2} x^2 dy$$

$$= \pi \int_0^{\pi/2} \cos^2 y dy$$

$$= \pi \int_0^{\pi/2} \frac{1}{2} (\cos 2y + 1) dy$$

$$= \frac{\pi}{2} \left[\frac{\sin 2y}{2} + y \right]_0^{\pi/2}$$

$$= \frac{\pi}{2} \left[\frac{\pi}{2} \right]$$

$$= \frac{\pi^2}{4} \text{ units}^3$$

(b) Let $u = 3 \sin x - 1$

$$\frac{du}{dx} = 3 \cos x$$

$$du = 3 \cos x \cdot dx$$

when $x = 0, u = -1$

$x = \pi/2, u = 2$

$$\int_0^{\pi/2} (3 \sin x - 1) \cos x dx = \frac{1}{3} \int_{-1}^2 u^2 du$$

$$= \frac{1}{9} [u^3]_{-1}^2$$

$$= \frac{1}{9} [8 + 1]$$

$$= 1$$

(c) (i) $\sin \theta = \frac{2t}{1+t^2}$

$$\cos \theta = \frac{1-t^2}{1+t^2}$$

(ii)

$$\sqrt{3} \sin \theta = 1 + \cos \theta \quad \text{--- (1)}$$

$$\sqrt{3} \times \frac{2t}{1+t^2} = 1 + \frac{1-t^2}{1+t^2}$$

$$\frac{2\sqrt{3}t}{1+t^2} = \frac{1+t^2+1-t^2}{1+t^2}$$

$$\therefore 2\sqrt{3}t = 2$$

$$t = \frac{1}{\sqrt{3}}$$

$$\tan \frac{\theta}{2} = \frac{1}{\sqrt{3}}$$

$$\frac{\theta}{2} = \frac{\pi}{6}$$

$$\therefore \theta = \frac{\pi}{3}$$

test $\theta = \pi$:

LHS of (1) $= \sqrt{3} \sin \pi = 0$

RHS of (1) $= 1 + \cos \pi = 0$

$\therefore$ solution are

$$\theta = \frac{\pi}{3} \text{ or } \pi$$

$$0 \leq \frac{\theta}{2} \leq \pi$$

Question 4

a) (i)

$$\frac{d}{dx} [x \tan x - \ln(\cos x)] = \frac{d}{dx} (x \tan x + \ln \cos x)$$

$$= \tan x + x \sec^2 x + \frac{1}{\cos x} \times (-\sin x)$$

$$= \tan x + x \sec^2 x - \tan x$$

$$= x \sec^2 x$$

$$= x \sec^2 x$$

(ii) $\int_0^{\pi/4} x \sec^2 x dx$

$$= \left[x \tan x - \ln \sec x \right]_0^{\pi/4}$$

$$= \frac{\pi}{4} \tan \frac{\pi}{4} - \ln \left(\frac{1}{\cos \frac{\pi}{4}} \right)$$

$$= 0 + \ln \left(\frac{1}{\cos 0} \right)$$

$$= \frac{\pi}{4} - \ln(\sqrt{2})$$

(b) (i) ${}^8C_4 = 70$

(ii) ${}^8C_6 = 28$

(iii) $\frac{{}^{10}C_6 - 70 - 28}{{}^{10}C_6}$

$$= \frac{112}{210}$$

$$= \frac{8}{15}$$

(c) $\ddot{x} = \frac{18}{(x-4)^2} \text{ m s}^{-2}$

(i) to the right

(ii) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{18}{(x-4)^2}$

$$\frac{1}{2} v^2 = \frac{18(x-4)^{-1}}{-1} + c$$

when $x=0, v=0$

$$\therefore 0 = \frac{-18 \times}{-4} + c$$

$$c = -\frac{18}{4}$$

$$\therefore \frac{1}{2} v^2 = \frac{-18}{x-4} - \frac{18}{4}$$

when $x=2$

$$\frac{1}{2} v^2 = \frac{-18}{-2} - \frac{18}{4}$$

$$= 4\frac{1}{2}$$

$$\therefore v = 3 \text{ m s}^{-1}$$

Question 5

$$x = 4 \sin \frac{\pi}{2} t$$

$$(i) \dot{x} = \frac{4\pi}{2} \cos \frac{\pi}{2} t$$

$$\ddot{x} = \frac{-4\pi^2}{4} \sin \frac{\pi}{2} t$$

$$= -\frac{\pi^2}{4} x$$

$$\Rightarrow \text{SHM, } n = \frac{\pi}{2}$$

$$(ii) \text{ amplitude} = 4$$

$$\text{period} = \frac{2\pi}{\pi/2} = 4$$

$$(iii) \text{ max speed is when } x = 0$$

$$x = 0$$

$$0 = 4 \sin \frac{\pi}{2} t$$

$$\frac{\pi}{2} t = \pi$$

$$t = 2$$

$$\dot{x} = \frac{4\pi}{2} \cos \pi$$

$$= -2\pi \text{ ms}^{-1}$$

$$\therefore \text{max speed} = 2\pi \text{ ms}^{-1}$$

(b) (i)

$$\begin{array}{r} x^2 - x + 4 \\ x^2 + 0x - 3 \overline{) x^4 - x^3 + x^2 - x + 1} \\ \underline{x^4 + 0x^3 - 3x^2} \\ -x^3 + 4x^2 - x \\ \underline{-x^3 + 0x^2 + 3x} \\ 4x^2 - 4x + 1 \\ \underline{4x^2 + 0x - 12} \\ -4x + 13 \end{array}$$

$$x^4 - x^3 + x^2 - x + 1 = (x^2 - 3)(x^2 - x + 4) - 4x + 13$$

$$(c) \frac{d}{dx} \cos^{-1}(e^{-x})$$

$$= \frac{-1}{\sqrt{1 - e^{-2x}}} \times (-1)(e^{-x})^2 \times e^{-x}$$

$$= \frac{e^x}{e^{2x} \sqrt{1 - e^{-2x}}}$$

$$= \frac{1}{e^x \sqrt{1 - e^{-2x}}}$$

$$= e^x \sqrt{\frac{e^{2x} - 1}{e^{2x}}}$$

$$= e^x \frac{1}{\sqrt{e^{2x} - 1}}$$

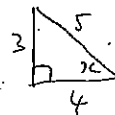
$$= \frac{1}{\sqrt{e^{2x} - 1}}$$

Question 6

a) At Pt of int:

$$4 \sin x = 3 \cos x$$

$$\tan x = \frac{3}{4}$$



let $x = \theta$ at pt of intersection

$$A = \int_0^{\theta} 4 \sin x \, dx + \int_{\theta}^{\pi/2} 3 \cos x \, dx$$

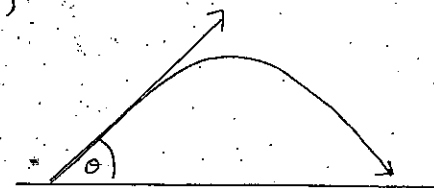
$$= 4 [\cos x]_0^{\theta} + 3 [\sin x]_{\theta}^{\pi/2}$$

$$= -4 [\cos \theta - \cos 0] + 3 [\sin \frac{\pi}{2} - \sin \theta]$$

$$= -4 \left[\frac{4}{5} - 1 \right] + 3 \left[1 - \frac{3}{5} \right]$$

$$= 2 \text{ m}^2$$

(b)



$$x = Vt \cos \theta$$

$$y = Vt \sin \theta - \frac{1}{2} g t^2$$

(i) Max height when $\dot{y} = 0$

$$\dot{y} = V \sin \theta - g t = 0$$

$$\text{when } t = \frac{V \sin \theta}{g}$$

$$\begin{aligned} \text{then } y &= V \sin \theta \times \frac{V \sin \theta}{g} - \frac{g}{2} \times \frac{V^2 \sin^2 \theta}{g^2} \\ &= \frac{V^2 \sin^2 \theta}{2g} \end{aligned}$$

(ii) $\dot{x} = V \cos \theta$ at max height

c)

$$\begin{aligned} 1) V &= \frac{1}{3} \pi R^2 H \\ &= \frac{1}{3} \pi \left(\frac{H}{2}\right)^2 \times H \\ &= \frac{1}{12} \pi H^3 \end{aligned}$$

$$i) \frac{dV}{dt} = 10$$

$$\frac{dh}{dt} = ? \text{ when } h = 50.$$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

$$\begin{aligned} \frac{dV}{dh} &= \frac{3}{12} \pi h^2 \\ &= \frac{1}{4} \pi h^2 \end{aligned}$$

$$\frac{dh}{dt} = \frac{4}{\pi h^2} \times 10$$

$$\begin{aligned} &= \frac{40}{\pi \times 2500} \text{ when } h = 50 \\ &= \frac{2}{125\pi} \end{aligned}$$

$$iii) \frac{dV}{dt} = -\pi h^2$$

$$\begin{aligned} \frac{dh}{dt} &= \frac{4}{\pi h^2} \times (-\pi h^2) \\ &= -4 \end{aligned}$$

$$iv) \text{ When } h = 0, t = ?$$

$$h = 100 \text{ when } t = 0$$

$$\frac{dt}{dh} = -\frac{1}{4}$$

$$t = -\frac{1}{4}h + c$$

$$\text{Sub } t = 0, h = 100$$

$$0 = -25 + c$$

$$c = 25$$

$$\therefore t = -\frac{1}{4}h + 25$$

Question 7

$$2) \frac{dM}{dt} = -kM$$

$$i) M = M_0 e^{-kt}$$

$$\begin{aligned} \frac{dM}{dt} &= -kM_0 e^{-kt} \\ &= -kM \end{aligned}$$

$$(ii) \text{ when } t = T$$

$$M = \frac{1}{2}M_0$$

$$-kT$$

$$\therefore \frac{1}{2}M_0 = M_0 e^{-kT}$$

$$\frac{1}{2} = e^{-kT}$$

$$\ln \frac{1}{2} = -kT$$

$$k = \frac{-\ln(\frac{1}{2})}{T}$$

$$= \frac{\ln 2}{T}$$

→ alternative:

$$\text{If } \frac{dh}{dt} = -4 \text{ mm/s}$$

To lower by 100 mm

at 4 mm/s takes 25s

b)

1) $\hat{P} = \theta$ (Lat Circump = $\frac{1}{2} \times$ Lat centre)
and $\angle XPO = \frac{\theta}{2}$

$$\therefore \hat{POX} = 180 - \theta \quad \left(\begin{array}{l} \text{is } \Delta XPO \\ \text{sum of } \Delta = 180 \end{array} \right)$$

$$A = \text{area } \triangle XOY + 2 \times \text{area } \triangle XPO$$

$$= \frac{1}{2} r^2 \sin 2\theta + 2 \times \frac{1}{2} r^2 \sin (180 - \theta)$$

$$= \frac{1}{2} r^2 \times 2 \sin \theta \cos \theta + r^2 \sin \theta$$

$$r^2 \sin \theta (\omega \sin \theta + i)$$

(11) $A = r^2 \sin \theta (1 + \cos \theta)$

$$\frac{dA}{d\theta} = r^2 \left[\cos\theta (1 + \cos\theta) + \sin\theta (-\sin\theta) \right]$$

$$= r^2 [\cos^2 \theta + \cos^2 \theta - \sin^2 \theta]$$

$$= r^2 [\cos \theta + \cos^2 \theta - (1 - \cos^2 \theta)]'$$

$$= r^2 [2 \cos^2 \theta + \cos \theta - 1]$$

(iii) For max area $\frac{dA}{dx} = 0$

$$2\cos^2 \theta + \cos \theta - 1 = 0$$

$$(2\cos\theta - 1)(\cos\theta + 1) = 0$$

$\therefore \cos \theta = \frac{1}{2}$ or $\cos \theta = -1$
✓ X

$$\therefore \theta = -60^\circ$$

$$\angle xpy = 60^\circ$$

Since $\triangle XPY$ is isosceles,
it must be equilateral.